TheWeatherMind

Project Team

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Chapter 1

Introduction

This document specifies the requirements of the application-TheWeatherMind. At a glance it is intended to be an AI-driven weather forecasting model, but it also includes other interesting features that will be discussed in detail in the following sections.

The document is intended for people directly involved in the industries of Agriculture, Disaster Management, Aviation, Climate Change, and Artificial Intelligence - specifically, developers, project managers, users, and CEOs.

1.1 Problem Statement

Weather forecasting is a major concern when it comes to Pakistan. It is crucial for industries such as aviation, disaster management, and agriculture. Existing models that we use on a daily basis like those of Google, and ECMWF only provide an accuracy up to 60%-70%. While these models are effective for other regions and countries, they face quite a few challenges when applied to Pakistan. They have been inadequately trained on the data from Pakistan which renders these models unreliable. Hence, the challenge to create a region-specific AI-model with improved accuracy remains untapped.

Our attempts with creating this model aim to address this gap by using localised weather data from **WeatherWalay** and advanced machine learning algorithms tailored specifically for the region of Pakistan. By doing so, we can provide accurate and precise weather predictions, reducing our reliance on the external data sources and models. This will not only save Pakistan the millions of rupees being spent on these sources but also allow us to have our own independent weather forecasting model.

1.2 Scope

We're aiming to build an AI-driven Weather Forecasting System focusing primarily on the regions of Islamabad, Lahore, and Karachi. This will include providing hyper-localised weather predictions down to the regional level. The project is divided into 4 major phases. **The first phase** of the project will consist of developing the core AI-driven weather model, trained on 3 different data sources: weather stations, radar data, and satellite imagery. The data from the radar will need a significant amount of pre-processing and will need to be converted to a format that would be usable. The AI-model will employ advanced machine learning algorithms resulting in highly localised and accurate weather predictions for the aforementioned regions. The reason for targeting these major cities of Pakistan is because WeatherWalay has the highest concentration of weather stations deployed in these areas. **The second phase** involves developing a user-friendly front-end interface for both web and mobile devices. This will provide a seamless and interactive view of the system for the users where they will be able to access the modules of weather predictions, weather news, and the purchase of weather data. **The third phase** focuses on creating a content generating module that uses open source models for generating text, audio, and video forecasts. This will be tailored to use real time data to provide the users with a personalised experience. To achieve this we will do prompt engineering based on the raw weather predictions to produce forecasts that are easy for the users to understand. **In the fourth phase** we will develop a blockchain-based transaction feature to enable secure and transparent transactions for users interested in purchasing localised weather data. This will provide a revenue model to support and expand the forecasting services.

1.3 Modules

The modules of our project includes training of model for accurate weather predictions, front-end design for web and mobile devices, dynamic weather news report generation, and blockchain based smart contracts for purchase of weather data.

1.3.1 Model training for Precise Weather Prediction

The focus of this module is to develop a machine learning model combining data from various data sources to predict weather variables.

1. Integration of data from three sources: weather station, radar, and satellite imagery
2. Pre-processing the data from radar
3. Model training for better weather prediction

4. Performance evaluation and optimization

1.3.2 Front-End Design for Mobile and Web

The second module is to design a user interface for mobile and web app to ensure a smooth user experience.

Client Web App

1. User authentication and authorization (optional)
2. User-friendly navigation
3. Weather Forecasts
4. Dynamic Weather News
5. Blockchain based transaction of weather data

Client Mobile App

1. Mobile optimised design for ease of use
2. Dynamic Weather Forecasts
3. Weather News

Admin Web App

1. User authentication and authorization
2. Payment verification
3. Audit trail
4. Dispute management

1.3.3 Weather News Reporting

Third module is to add on a feature to deliver dynamic weather news reports in multiple formats providing personalised experience to the users

1. Prompt engineering based text generation from raw weather data

User class	Description
Weather Enthusiasts	A weather enthusiast is a person who is interested in weather updates for personal use, like planning the daily activities or trips. The visuals of app will be simple and easy for all type of Weather Enthusiasts. An estimated 70 percent of Weather Enthusiasts will use the mobile app an average of 3-5 times per day.
Agriculture Professionals	Agriculture professionals are people who are either agriculture consultants or farmers. They rely on accurate weather predictions for decision making in agricultural operations like irrigation and harvesting schedules. Agricultural professionals are more interested in detailed weather predictions.
City Planners and Government Officials	City Planners and Government Officials require weather predictions to take necessary steps for severe conditions like heavy rainfall or storms. This class of users are likely to access the weather forecast service through desktop web application
Weather Data Purchasers	Weather Data Purchasers are companies, organizations, or even individuals who are interested in purchasing localized weather data for commercial purposes. Such users will use blockchain based transcation feature for the purchase of the data and they are likely to use web application.
Weather Reporters	Weather Reporters include people who prodcue weather related content for public. This class may include new reporters, influencers, or news organizations. These type of users will access the feature of dynamic news reports via mobile or web application both.

2. Text-to-audio generation for weather updates in professional voice, adjusting tone and speed
3. Audio-to-video generation where an avatar will be narrating new updates, with lip sync and synchronized captions

1.3.4 Blockchain Based Transactions - Smart Contracts

The last module is to integrate blockchain to ensure secure and transparent exchange of weather data.

1. Implementation of smart contracts for purchase of weather data
2. Payment verification feature to ensure verified payments before providing data access to the customer
3. Automated data release will enable automatic access to the customer once the payment is verified
4. Audit trail to keep purchase and exchange of data transparent to both parties
5. Dispute Management in case of failed transactions or refund scenarios

1.4 User Classes and Characteristics

Chapter 2

Project Requirements

2.1 Event Response Table

Event	Trigger	Response
User requests weather forecast	User selects the desired location	System retrieves weather data from backend (AI predictions, weather station data) and displays results.
New radar data received	New radar data is appended in the database	System pre-processes the radar data, and feeds it for the training of the AI model.
User changes the location for the weather forecast	User selects a new location from the drop down	System fetches weather data for the selected location and displays updated forecast.
User requests to view weather news	User selects "View Weather News" feature	System generates a video report based on real-time data through text and with synchronised captions.
User selects location and weather variables for purchase	User selects up to 5 variables (e.g., temperature, humidity, precipitation)	System creates a data package according to the selected variables and calculates the price for the user.
Purchase process initiated	User confirms a purchase of weather data	System processes payment through blockchain and grants access to a downloadable file containing the data.

Table 2.1: Event-Response Table

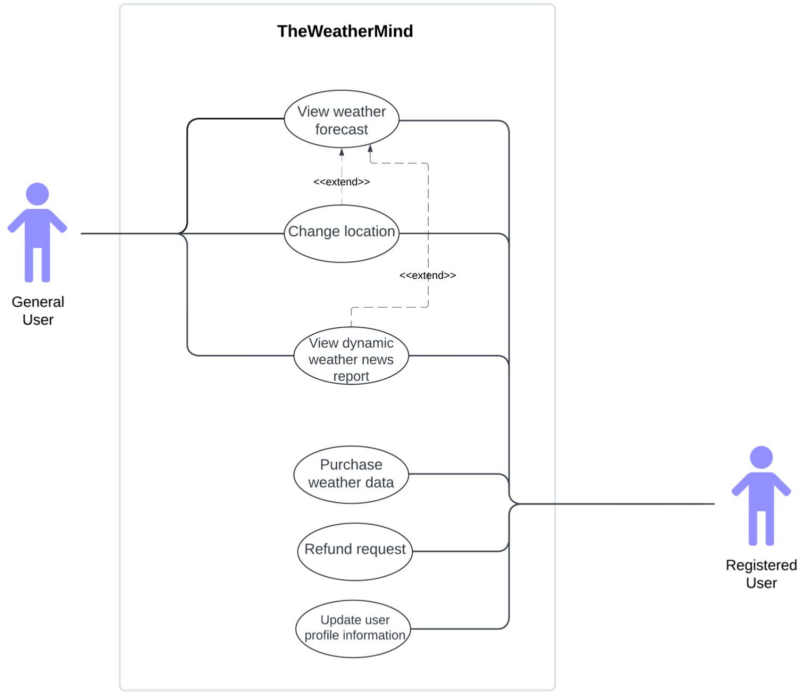

Figure 2.1: Use Case Diagram

Use Case Name	Change Location
Scope	TheWeatherMind
Level	User level
Primary Actors	General and Registered User
Stakeholders and Interests	**General User:** General user wants real-time and accurate weather forecast for specific location. **Weather Data Providers:** Weather data providers provide the weather data required for weather forecasting.
Preconditions	1. User has launched the app. 2. App can access the user's location. 3. Weather data services are available and functional. 4. User has access to a stable internet connection.
Postconditions	The user successfully views the nowcast and the forecast for the new location that they have selected.

Main Success Scenario	**Actor Action**	**System Response**
	1. User selects the "change location" option	2. System provides a drop down to select the desired location.
	3. User selects the desired location in the input field.	4. System fetches weather information for the specific location from the back-end.
	5. User views the weather forecast	6. System provides a user-friendly display of weather forecast for new location.
Alternative Flows	**Actor Action**	**System Response**
	1a. User inputs invalid location.	2a. System displays error message that the entered location is not supported or not recognized.

Table 2.3: Change Location Use Case

Use Case Name	View Dynamic Weather News Reports
Scope	TheWeatherMind
Level	User level
Primary Actors	General and Registered User
Stakeholders and Interests	**General User:** General user wants real-time weather news with audio, video, and captions for a better experience. **Weather Data Providers:** Weather data providers provide the weather data required for weather forecasting.
Preconditions	1. User has launched the app. 2. App can access the user's location. 3. Weather data services are available and functional. 4. User has access to a stable internet connection.
Postconditions	The user successfully views the dynamic weather news with a seamless experience.

Main Success Scenario	**Actor Action**	**System Response**
	1. User selects the "dynamic weather news report" option	2. The system displays dynamic news report with seamless video, clear audio, and synchronized captions.
	3. User views the dynamic news reports.	4. System continues to provide uninterrupted video and quality audio and captions.
Alternative Flows	**Actor Action**	**System Response**
	1a. User has slow or unstable internet connection.	2a. System detects the slow connectivity and buffers the video.
	1b. User selects "dynamic news reports" option but news content is not available.	2b. System displays that the video is currently not available and offers an alternative summary of the news report.

Table 2.4: View Dynamic Weather News Report

Use Case Name	Purchase Weather Data
Scope	TheWeatherMind
Level	User level
Primary Actors	Registered User
Stakeholders and Interests	**Registered User:** Registered user wants to purchase the weather data using blockchain-based smart contracts. **WeatherWalay:** Weatherwalay want to sell the data of their weather stations on user's request.
Preconditions	1. User has registered for the premium web app features. 2. User has logged into the account on the web app. 3. Weather data is available to provide the customer when they make a request. 4. Blockchain payment systems are operational for secure delivery and payment of data.
Postconditions	The user has successfully purchased the weather data and gets access to download it, while the payment and transaction history is successfully recorded and logged.

	Actor Action	System Response
Main Success Scenario	1. User navigates to "Purchase Weather Data" feature and selects the desired location.	2. The system displays different weather parameters (e.g., temperature, humidity, precipitation) to add to the downloadable file.
	3. User selects upto 5 weather parameters that they want.	4. System creates and displays the data package's details and pricing along with the prompt to continue to payment.
	5. User initiates the payment process.	6. System prepares the smart contract for the transaction.
	7. User confirms the payment.	8. System processes the payment via blockchain and confirms the successful transaction.
	9. User receives confirmation of payment and access to the weather data purchased.	10. System confirms the completion of the purchase made, and saves the records of the purchase and transaction.

<table>
<tr><td rowspan="4">Alternative Flows</td><td>Actor Action</td><td>System Response</td></tr>
<tr><td>3a. User selected weather parameters which are not available for the specific location.</td><td>4a. System displays an error message stating the selected parameter is not available at the moment, try again later or we will notify you when the data is available.</td></tr>
<tr><td>5a. User tries to initiate the payment process but the blockchain payment service is down.</td><td>6a. System displays an error message stating the system is down or the service is currently not available.</td></tr>
<tr><td>7a. User decides to cancel the purchase in the middle of the payment process.</td><td>8a. System cancels the transaction and returns the user to the weather data selection page.</td></tr>
</table>

Table 2.5: Purchase Weather Data

Use Case Name	Refund Request
Scope	TheWeatherMind
Level	User level
Primary Actors	Registered User
Stakeholders and Interests	**Registered User:** Registered user wants to receive a refund for purchased weather data. **Blockchain Gateway:** Ensures that all refund transactions are processed securely and transparently.
Preconditions	1. User has logged in to the web app. 2. User has already purchased the weather data. 3. The refund request they are trying to make is within the allowed time frame.
Postconditions	The user successfully receives the refund and the payment records are updated.

	Actor Action	System Response
Main Success Scenario	1. User navigates to the "Purchase History" and selects the purchase they want a refund for.	2. System displays history of user's purchases and displays the refund option where applicable.
	3. User clicks on the "Request Refund" button.	4. System confirms that refund is valid and displays a refund form for user confirmation.
	5. User fills the form and submits the refund request.	6. The blockchain-based transaction system updates the purchase status to "Refund Pending".
		7. System processes the refund, and updates the purchase status to "Refund Completed".
	8. User receives the refund successfully.	9. System records refund and updates the purchase history successfully.
Alternative Flows	**Actor Action**	**System Response**
	3a. User tries to request a refund but it falls outside the allowed time frame.	4a. System displays an error message that "Refund is not applicable".
	3b. User starts the refund process but cancels it before successful submission.	4b. System cancels the refund process and returns the user to the purchase history page.
	5a. User submits the refund request but the transaction system is down or not working.	6a. System displays an error message that "Service is down or currently not available".

Table 2.6: Refund Request

<table>
<tr><td>Use Case Name</td><td>View Weather Forecast</td></tr>
<tr><td>Scope</td><td>TheWeatherMind</td></tr>
<tr><td>Level</td><td>User level</td></tr>
<tr><td>Primary Actors</td><td>General and Registered User</td></tr>
<tr><td>Stakeholders and Interests</td><td>General User: General user wants real-time and accurate weather forecast.
Weather Data Providers: Weather data providers provide the weather data required for weather forecasting.</td></tr>
<tr><td>Preconditions</td><td>1. User has launched the app.
2. App can access the user’s location.
3. Weather data services are available and functional.</td></tr>
<tr><td>Postconditions</td><td>The user successfully views the nowcast and the forecast of the following hours.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><td>Actor Action</td><td>System Response</td></tr>
<tr><td>1. User opens the app</td><td>2. System detects user location.</td></tr>
<tr><td></td><td>3. System fetches weather information for the specific location from the back-end.</td></tr>
<tr><td>4. User views the weather forecast</td><td>5. System provides user-friendly display of weather forecast.</td></tr>
</table></td></tr>
<tr><td>Alternative Flows</td><td><table>
<tr><td>Actor Action</td><td>System Response</td></tr>
<tr><td>1a. User opens app with no internet connection.</td><td>2a. System displays error message that weather information can not be fetched without internet connection.</td></tr>
<tr><td>1b. User opens app with location services disabled.</td><td>2b. System displays message to enable location services to fetch weather forecast.</td></tr>
</table></td></tr>
</table>

Table 2.2: View Weather Forecast

Use Case Name	Update User Profile Information
Scope	TheWeatherMind
Level	User level
Primary Actors	Registered User
Stakeholders and Interests	**Registered User:** Registered user wants to update their profile information and keep their information up to date.
Preconditions	1. The user already has a profile. 2. User has logged into the web app successfully.
Postconditions	User has successfully made the changes in the profile information and the system updates the user information.

	Actor Action	System Response
Main Success Scenario	1. User navigates to the "Account Settings" section.	2. The system displays current profile information of the user.
	3. User clicks on the "Edit" button.	4. System enables the editable fields of the profile.
	5. User updates the fields with new information.	6. System validates the entered information (e.g., syntax).
	7. User confirms the updates by clicking on the "Save Changes" button.	8. System updates the user profile in the records.
	9. User gets an alert message "Profile updated Successfully".	
Alternative Flows	**Actor Action**	**System Response**
	5a. User inputs invalid data in the input fields (e.g., incorrect syntax).	6a. System displays an error message "Incorrect input" and prompts the user to enter correct information.
	7a. User decides to cancel the profile update before saving the changes.	8a. System displays an error message "Your changes are not saved" and returns the user to the "Account Settings" page.
	7b. User clicks on "Save Changes" button, but a network issue prevents updating the profile information.	8b. System displays a message that "Changes not saved, try again later".

Table 2.7: Update User Profile Information

2.2 Functional Requirements

2.2.1 Weather Prediction Engine

Following are the requirements for module 1:

1. The system shall collect real-time data from weather stations and radar
2. The AI model shall process and analyze the data to generate accurate localized weather forecasts
3. The model shall provide predictions for specific geographical locations (e.g., Islamabad, Lahore, Karachi)
4. The system shall update weather forecasts at hourly intervals
5. The model shall allow for continuous training and improvement as new data is collected

2.2.2 Forecast Interface

Following are the requirements for module 2:

1. The front-end shall display real-time weather forecasts, including temperature, precipitation, humidity, dew, and other relevant metrics
2. Users shall be able to select specific locations to view localized weather predictions
3. The design shall incorporate weather updates in textual, audio, and video formats
4. The design shall integrate blockchain-based transactions for users who wish to purchase localized weather data

2.2.3 Weather Content Generator

Following are the requirements for module 3:

1. The system shall generate weather forecasts in text, audio, and video formats using real-time data
2. Text prompts shall be automatically generated based on weather data to create human-readable content

3. The audio and video generation shall provide accurate spoken forecasts using natural language processing (NLP) techniques

4. The module shall allow users to switch between different content formats (text, audio, video)

2.2.4 Transaction System

Following are the requirements for module 4:

1. The system shall allow users to securely purchase localized weather data using blockchain technology

2. The blockchain shall record every transaction for weather data purchases in an immutable ledger

3. Users shall be able to verify the authenticity of weather data transactions through blockchain

4. The system shall provide users with a seamless and transparent process for data transactions

5. The transaction system shall support payments in local currency

6. The system shall notify users once transactions are successfully completed

2.3 Non-Functional Requirements

2.3.1 Reliability

TheWeatherMind system shall be designed to minimise failures, aiming for a Mean Time Between Failures (MTBF) of at least 600 hours (25 days). A failure is defined as any instance where the system provides weather forecast but it is not accurate enough or fails to display information correctly. To protect against failures, the system shall implement monitoring and diagnostics for proactive issue identification. Error detection strategies will include logging and alerting mechanisms, while error correction strategies will involve automated recovery processes to restore functionality swiftly.

2.3.2 Usability

USE-1: The system shall allow a user to access the localized weather forecast for their selected location within **two clicks** from the homepage.

USE-2: The interface shall provide clear visual feedback on user actions, such as loading indicators when fetching weather data, to minimize uncertainty during interactions.

USE-3: The system shall enable the users to seamlessly switch to weather news screen for weather reports and generate a video within **a minute.**

USE-4: The application shall allow users to search for weather data for any specified location within **7 seconds**.

USE-5: Once the location is selected, users can choose up to **5 specific weather parameters** (e.g., temperature, precipitation, wind speed) to include in a downloadable file, and finalize the selection within **3 clicks**.

USE-6: The application shall facilitate the purchase of weather data using a secure blockchain-based payment system, processing each transaction and providing a confirmation within **10 seconds** of payment initiation, while ensuring transparent and immutable transaction records.

2.3.3 Performance

PER-1: 95% of weather forecast data shall be generated and displayed to the user within **7 seconds** of the user's request, ensuring timely access to localized weather information.

PER-2: The system shall support at least **100 concurrent users** accessing weather data without degradation in performance, maintaining response time under **7 seconds** per request.

PER-3: Real-time updates to weather forecasts shall be processed and delivered within **1 minute** of data collection from weather stations, ensuring users receive the latest information.

PER-4: The system shall allow for weather data retrieval from the database with a maximum latency of **5 seconds** under normal operating conditions.

2.3.4 Security

Robust security measures shall protect sensitive data and ensure user privacy in TheWeatherMind system. Security requirements shall focus on preventing unauthorised access and safeguarding the integrity of weather data. The skill level and effort required to breach the system will be assessed, with regular security evaluations conducted to identify vulnerabilities and enhance protective measures without specifying particular solutions in this document.

2.3.5 Scalability

SCL-1: The system shall support up to **100** simultaneous users without degradation in performance (e.g., no significant delay in generating weather forecasts or processing transactions).

SCL-2: The infrastructure shall allow for horizontal scaling by adding more servers to accommodate growing user traffic, ensuring response time for data requests remain under **5 seconds.**

2.3.6 Portability

PRT-1: The system shall be accessible from various platforms, including web browsers, mobile applications (iOS and Android), and desktop applications without requiring significant changes to the underlying code.

PRT-2: The system's weather forecasting and data purchase features shall work seamlessly on all modern web browsers (Chrome, Firefox, Safari, and Edge) and mobile platforms with minimal differences in appearance or performance.

2.3.7 Extensibility

EXT-1: The system shall allow the integration of new weather data sources or third-party APIs with minimal modification to the existing architecture.

2.4 Domain Model

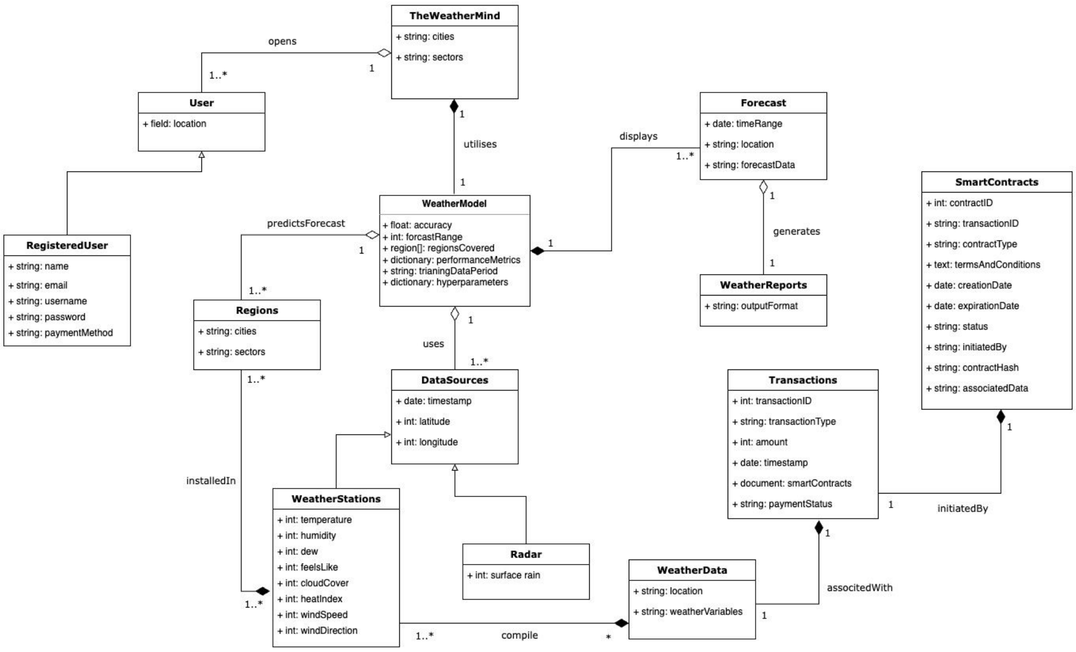

Figure 2.2: Domain Model

Chapter 3

System Overview

The **WeatherMind** AI-based weather forecast model aims to deliver accurate, hyper-localized weather predictions for major urban areas in Pakistan, including Islamabad, Lahore, and Karachi. By leveraging advanced machine learning algorithms and integrating data from multiple sources—such as weather stations, radar, and satellite imagery—the system provides real-time weather forecasts that users can customize by changing their locations. In addition to forecasting, WeatherMind features dynamic news reporting in text, audio, and video formats, enhancing user engagement and information delivery. Furthermore, registered users can purchase specific weather data, with transactions secured through blockchain technology to ensure transparency and trust. This innovative approach addresses the increasing demand for precise weather information, ultimately improving decision-making in various sectors affected by climate conditions. Provide any background information if necessary.

3.1 Architectural Design

3.1.1 Box and Line Diagram

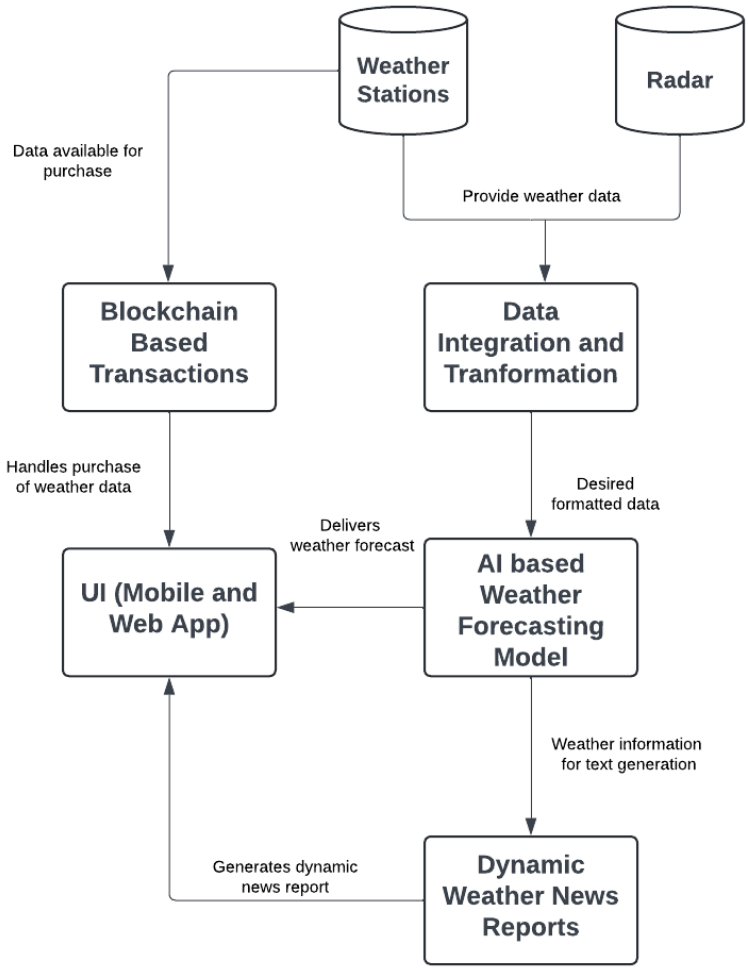

Figure 3.1: Box and Line Diagram

3.1.2 Architectural Diagram

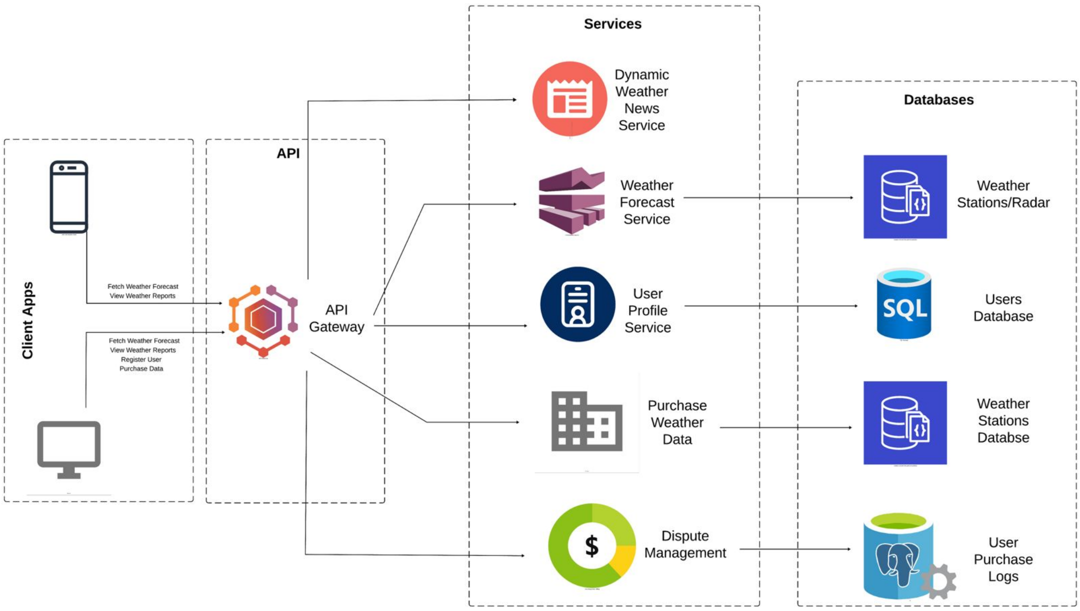

Figure 3.2: Architectural Diagram

3.2 Design Models

Object Oriented Development Approach

3.2.1 Activity Diagram

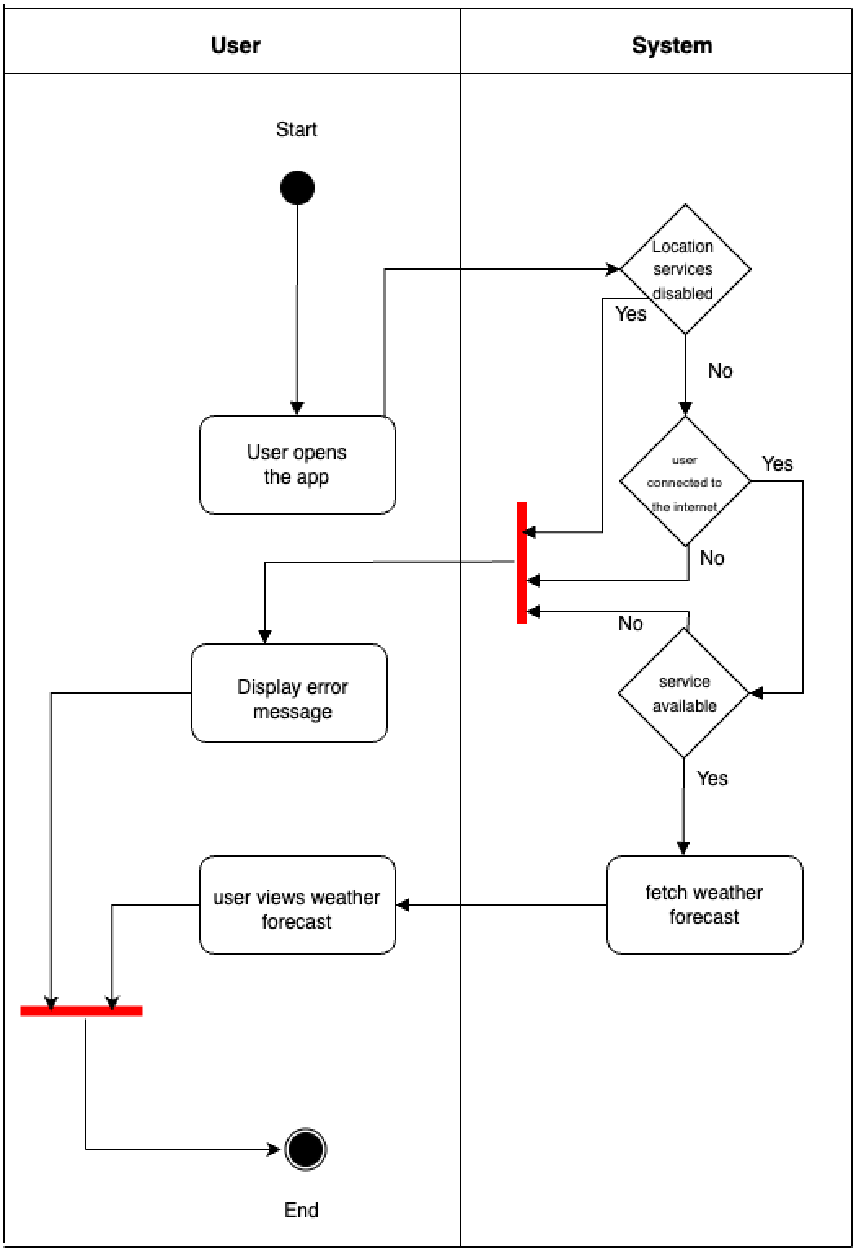

Figure 3.3: View Weather Forecast

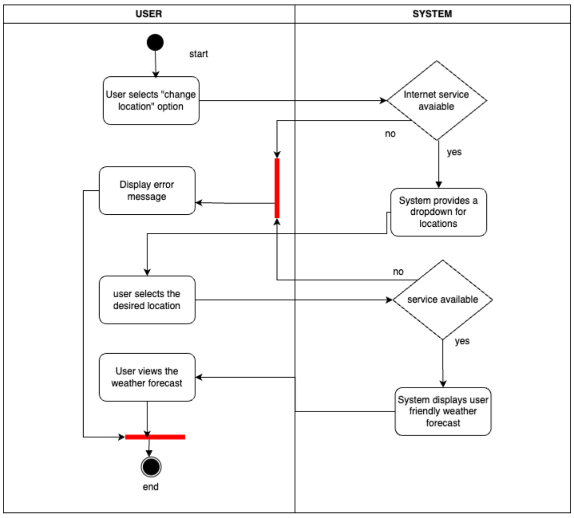

Figure 3.4: Change Location

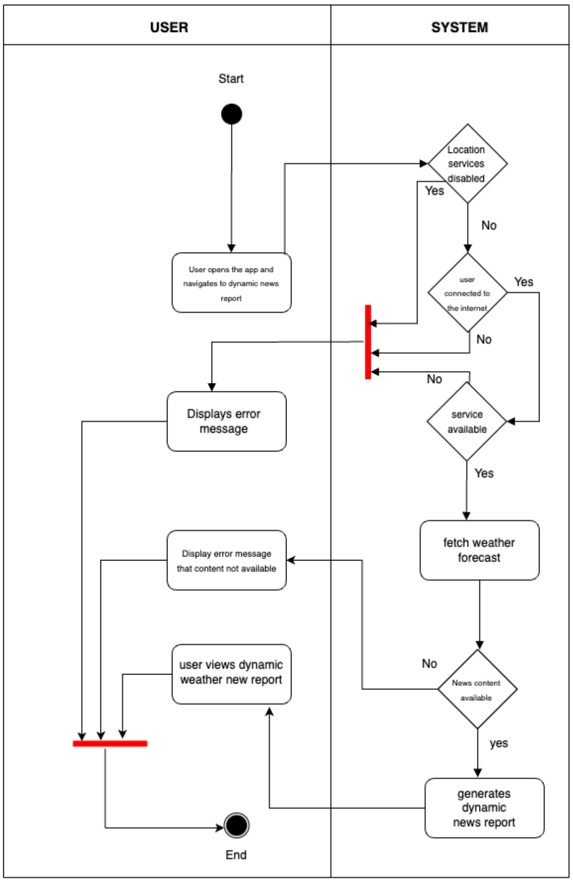

Figure 3.5: Dynamic Weather News Report

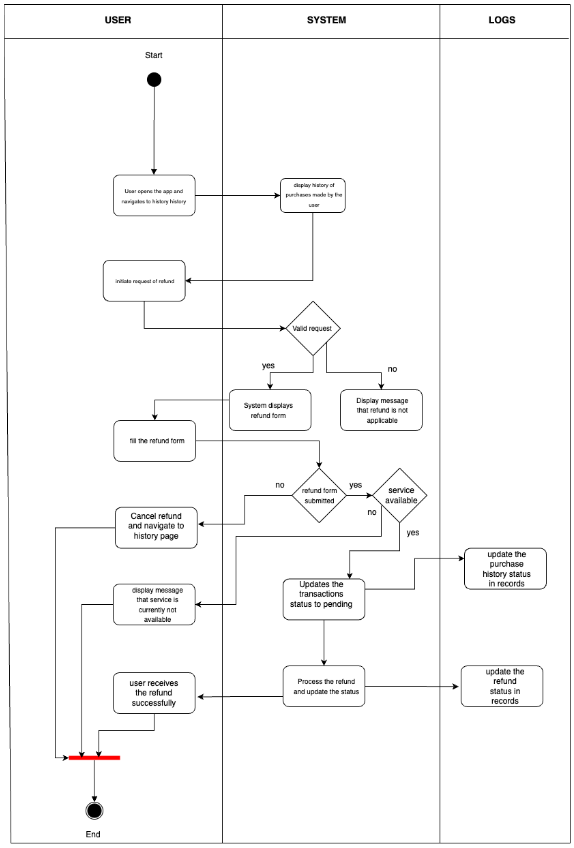

Figure 3.6: Purchase Weather Data

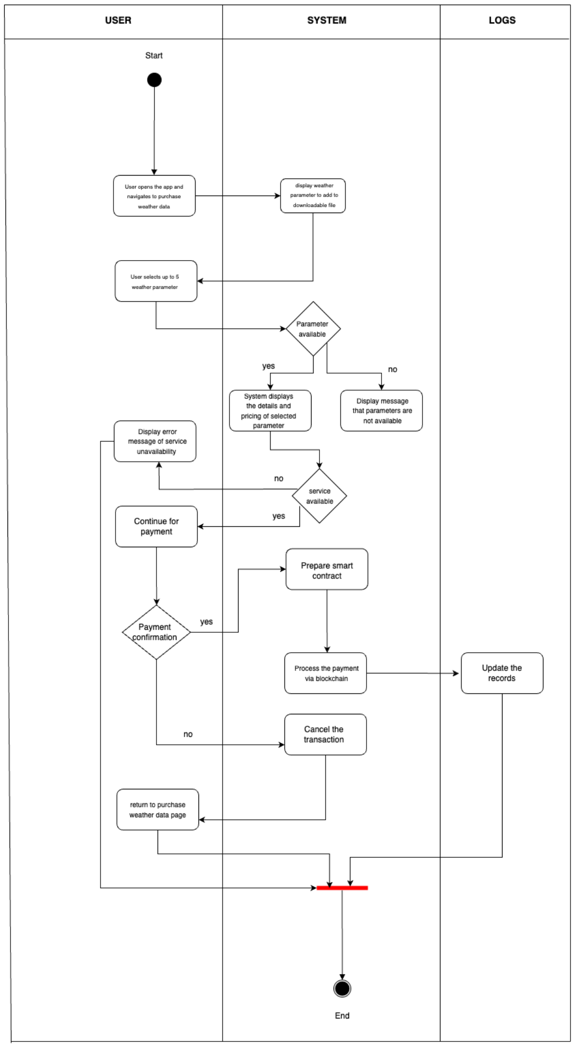

Figure 3.7: Refund Request

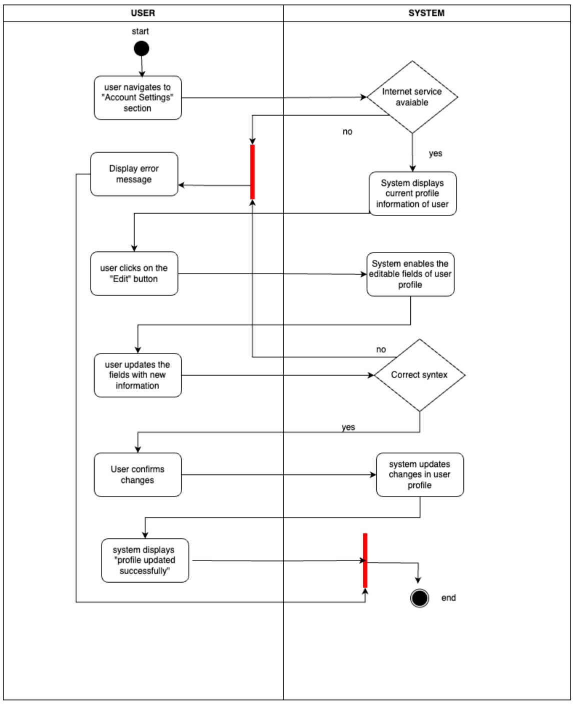

Figure 3.8: Update User Profile Information

3.2.2 Class Diagram

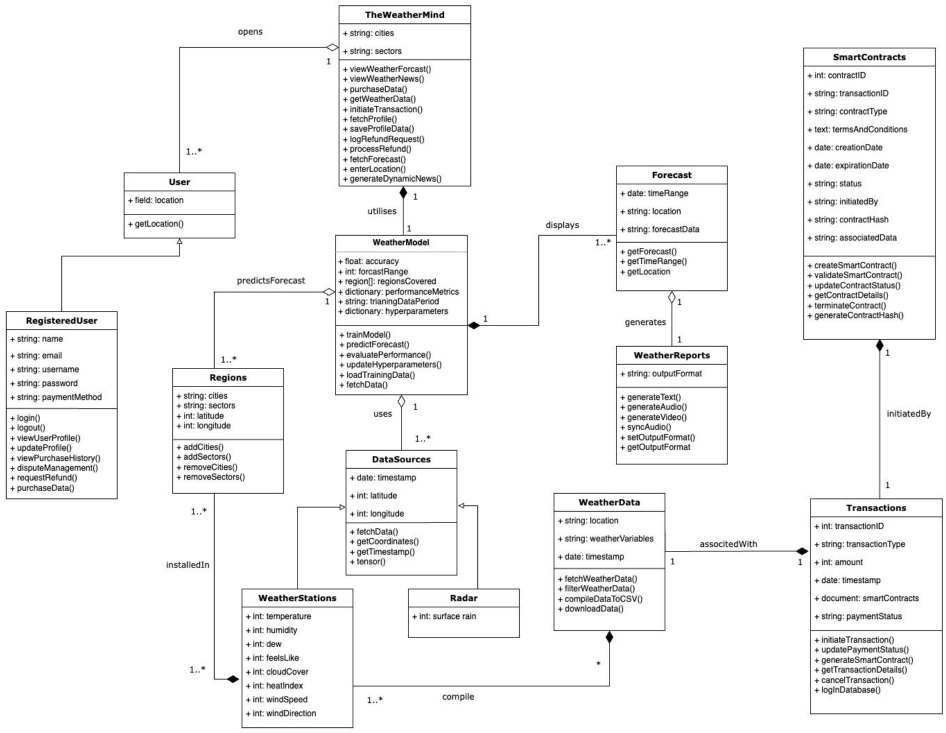

Figure 3.9: Class Diagram

3.2.3 Sequence Diagrams

Figure 3.10: View Weather Forecast

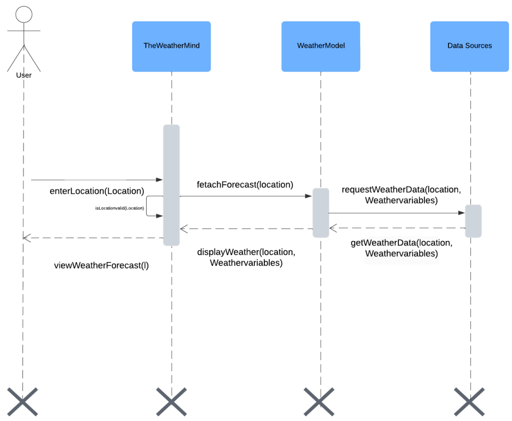

Figure 3.11: Change Location

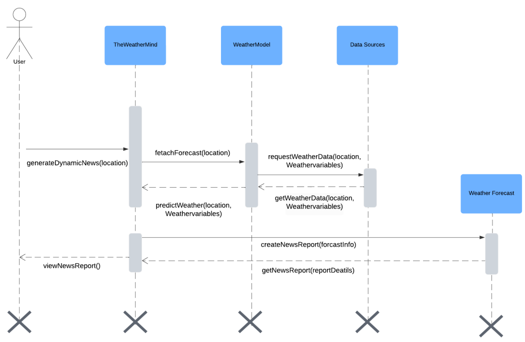

Figure 3.12: Dynamic Weather News Report

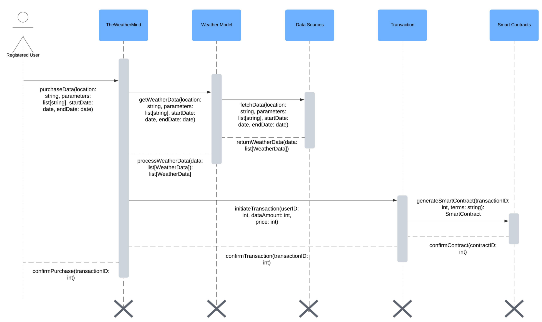

Figure 3.13: Purchase Weather Data

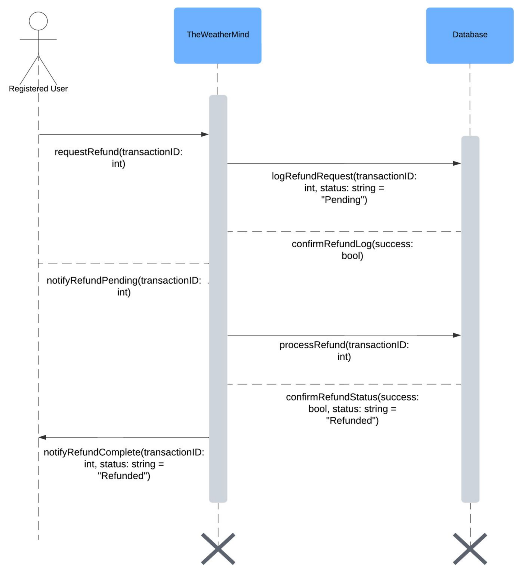

Figure 3.14: Refund Request

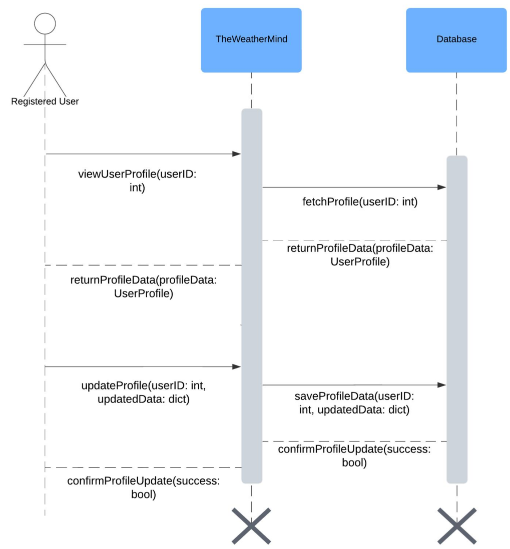

Figure 3.15: Update User Profile Information

3.2.4 State Transition Diagram

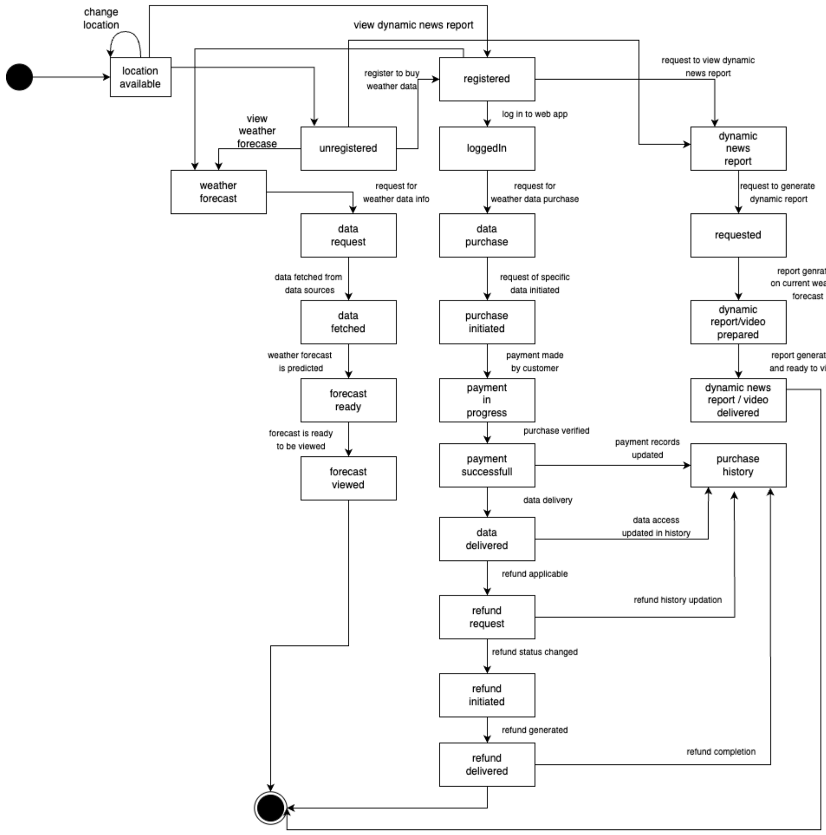

Figure 3.16: State Transition Diagram

3.3 Data Design

3.3.1 Introduction

The data design of TheWeatherMind focuses on how the weather data, radar data, user information, and transaction logs are stored, processed, and retrieved. The system utilises MongoDB for the storage of non-relational data and structured data like JSON to represent entities such as weather data, radar data, user information, and transaction logs.

3.3.2 Data Structures

The main data entities within TheWeatherMind are represented as collections in MongoDB. Each collection holds the data in the form of JSON that allows for easy and fast data retrieval. Key data structures used are the following:

- **Collections:** MongoDB stores data in collections. Each collection holds data representing different sources and different types of data. This includes data from the radar, data from the weather stations (Historic Weather data, Recent Stats), user information, and transaction logs.

- **Dictionaries:** Within these collections, weather data, and user information are stored as dictionaries, with key-value pairs mapping attributes like temperature, humidity, and timestamps.

An example of how our weather data is being stored is shown below:


```
{
  "_id": "662b464a208393ef7d71f4f7",
  "stationID": "25591",
  "__v": 0,
  "location": {
    "long": 73.39999914
  },
  "nowcast": {
    "time": "2024-10-06T22:00:00+0500",
    "temperature": -0.4,
    "humidity": 77,
    "windSpeed": 6
  }
}
```


- **Arrays:** Arrays are used to store collections of related data, such as daily and hourly weather records. Each array contains multiple entries, each representing a specific time period or data point (eg., hourly weather reports, daily weather summary).

3.3.3 Data Storage and Organisation

TheWeatherMind system organises its data into multiple MongoDB collections, each designed to handle specific entities within the system. The collections that are created are flexible, allowing for schema-less data storage, which is ideal for handling various forms of weather, radar, and user data.

- **Weather Data Collection:** Stores weather station data, including nowcast (real-time weather data), and hourly forecasts. Each document in this collection represents a weather station and contains nested fields for parameters like temperature, humidity, wind speed, cloud cover, dew and more.

  Stored Information:

- **stationID:** Identifies the weather station.
- **location:** Contains latitude and longitude data for the station.
- **nowcast:** Provides real-time data such as temperature, humidity, wind speed, and wind direction.

- **Radar Data Collection:** Stores radar data, including surface rain measurements and timestamps. This data is used in conjunction with weather station data to provide accurate real-time forecasts.

  Stored Information:

- **lat/long:** Latitude and longitude for the radar location.
- **surface rain:** Amount of surface rain detected by the radar.
- **timestamp:** Time of data capture.

- **User Information:** Stores user profiles and authentication information.

  Stored Information:

- **user id:** Unique identifier for the user.
- **profile details:** Name, email, and location.
- **Transaction Logs:** Stores transaction history for user purchases, refunds, and smart contracts.

  Stored Information:

- **transaction id:** Unique identifier for each transaction.
- **user id:** Identifier for the user making the purchase.
- **data purchased:** Weather data details, including location, parameters, and date range.
- **status:** Tracks whether the transaction is pending, completed, or refunded.
- **smartContracts:** Stores the contract associated with each transaction.

3.3.4 Data Preporcessing

The system processes the data stored in the MongoDB collections based on user requests, such as weather data retrieval, profile updates, and transactions.

3.3.4.1 Weather Data Retrieval

When a user requests weather data, the system fetches the appropriate data from the Weather Data Collection based on the user's selected parameters (e.g., temperature, wind speed, humidity) and location. Data is filtered by timestamp to meet the user's specific time range.

3.3.4.2 Radar Data Processing

Radar data is retrieved from the Radar Data Collection and used to generate real-time precipitation forecasts. This data is combined with weather station data to improve the accuracy of nowcasts.

3.3.4.3 User Profile Management

User profiles, stored in the User Collection, can be updated when users change their preferences or personal details. The system processes profile updates and stores the changes back in the MongoDB collection.

3.3.4.4 Transaction Handling

The system logs all purchases in the Transaction Logs Collection. Each purchase is associated with a smart contract, which is also stored in the same collection. The status of each transaction (e.g., pending, completed, refunded) is tracked and updated as the purchase progresses.

Bibliography